
Incentive Allocation for Fitbit Data Collection with Two-Stage Predict-Then-Optimize and Decision-Focused Learning

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Abstract

TODO: Briefly summarize the allocation problem, the methods compared, the main experimental setup, and the key findings.

1. Introduction

Wearable devices such as Fitbit and Apple Watch are increasingly used to collect longitudinal health behavior data for large-scale studies. Compared to traditional medical devices, they are relatively inexpensive and easy to deploy, making them attractive tools for population-level health monitoring and behavioral research. However, maintaining long-term participant compliance remains a major challenge.

In the ALiR study [cite], wearable devices were distributed to more than 1,000 participants who agreed to wear the device for at least 60% of the time over a one-year study period. In practice, only about one third of participants consistently met this compliance threshold. Moreover, while the initial participant population was balanced across demographic groups, some groups exhibited substantially higher dropout rates over time, leading to imbalance in the collected dataset.

Both reminders and monetary incentives may help improve participation. Survey responses from the ALiR study suggest that some participants simply forget to wear the device, indicating that reminder-based interventions may increase compliance. Monetary incentives may also improve adherence: during the first 12 months of the study, participants received a \$5 reward for each month in which they satisfied the compliance threshold, and participation declined after incentives ended.

These settings naturally lead to a contextual stochastic optimization problem. Before the next month begins, we ob-

serve contextual information such as demographic covariates and historical wear patterns, then decide which users should receive an intervention. The true compliance outcome is only realized afterward, introducing uncertainty into both the utility and cost of the decision.

A standard approach is the predict-then-optimize (PTO) framework, which first predicts future outcomes and then solves a downstream allocation problem using the predictions. However, in our setting, the downstream objective includes threshold utility, fairness considerations, and budget constraints, making accurate prediction alone insufficient for good decision quality. This motivates the use of decision-focused learning (DFL), where the predictive model is trained directly through the downstream optimization objective.

In this project, we study both ranking-based nudging interventions and knapsack-style monetary incentive allocation. We compare PTO and several DFL approaches, including end-to-end differentiable approximation, randomized smoothing, and perturbation-gradient methods, under threshold utility, fairness penalties, and stochastic budget constraints.

2. Related Work

Markov Behavior Model Markov models are frequently used to model behaviors [cite].

Treatment Allocation under Uncertain Response

Two-Stage Decision-Blind Methods

Decision-Focused Learning The key to decision-focused learning is how to design a full training pipeline so that we can directly optimize for the decision. One common approach is to train a neural network that takes contextual information as input and train it to output the optimal decision. The nontrivial components of this pipeline include 1) dealing with the non-differentiable parts present in the optimization problem and 2) guaranteeing feasibility. For feasibility, the easiest way is to have a fix routine that ensures feasibility as the last step

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of the pipeline. For non-differentiability, besides having no gradient at certain areas of the function, it also leads to noninformative gradient. Methods to deal with this include parametrization, randomized smoothing and perturbation gradient methods.

3. Approach

3.1. Simulation Setup

TODO: Describe the synthetic data generation process, potential outcomes, user features, and train/validation/test split.

3.2. Predict-Then-Optimize

TODO: Describe the supervised outcome model and downstream allocation rule.

3.3. Decision-Focused Learning

TODO: Describe the differentiable surrogate objective and how it is trained.

4. Results

4.1. Experimental Setup

All methods use the same outcome model architecture: a two-hidden-layer MLP with 64 ReLU units per layer and dropout 0.1. The input contains the group indicator, five baseline covariates, 30 pre-intervention wear indicators, and the action indicator, for 37 features total. Thus PTO, DFL, randomized smoothing, and perturbation-gradient methods have the same model capacity and differ only in their training objective and allocation surrogate.

For PTO, we use 5-fold cross-validation to choose the number of training epochs, then retrain the final model on the full training set for the mean best epoch. The DFL variants use the same protocol: each variant is cross-validated to select an epoch count and then retrained on all training data. All reported policies are evaluated on the same held-out test set under the same threshold, budget, and allocation setting.

5. Discussion

TODO: Interpret the results, note limitations, and describe what the findings imply for the allocation setting.

5.1. Presentation and Code Review Feedback

TODO: Explain what feedback was received during the presentation and code review, and how it was incorporated into the final project.